

Core Banking Services: Business Requirements

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Owner	Retail Banking Product Office
Approved by	Head of Retail Products; Head of Technology Risk
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Related documents	NBK-NFR-2026-003 (Non-functional requirements), NBK-SEC-2025-021 (Authentication standard)

1. Purpose and scope

This document states the functional requirements for the Accounts, Transfers, Loans and Security services of the Northbridge core banking platform. Every requirement has a unique identifier of the form **REQ-<MODULE>-<NN>**. Each identifier must be traceable to its implementation and to at least one automated test before a release is approved (see MAS Technology Risk Management Guidelines, section 9).

All monetary amounts are in Singapore Dollars (SGD) and are held to two decimal places. "Calendar day" means a day in Singapore Standard Time (UTC+8). Non-functional requirements (performance, availability, security controls) are out of scope and are covered in NBK-NFR-2026-003.

2. Glossary

Term	Meaning
Active account	An account in status ACTIVE. Accounts may also be FROZEN or CLOSED.
Payee	The destination account of a transfer.
New payee	A destination to which the source account has never completed a transfer.
Instalment	The fixed monthly repayment for an amortising loan.

3. Accounts module

REQ-ACC-01 Account opening

Requirement	A customer can open an account by giving their full name, an initial deposit and a 6-digit PIN.
Acceptance criteria	• A blank or whitespace-only name is rejected. • A negative initial deposit is rejected. An initial deposit of SGD 0.00 is allowed. • A newly opened account has status ACTIVE and a balance equal to the initial deposit.

REQ-ACC-02 Account number format

Requirement	Each account gets an account number made of the prefix "SG" and 8 digits, e.g. SG01234567.
Acceptance criteria	• Account numbers match the pattern <code>SG\d{8}</code>. • No two accounts share an account number.

REQ-ACC-03 Deposits

Requirement	A customer can deposit money into an active account.
Acceptance criteria	• The deposit amount must be greater than zero; zero or negative amounts are rejected. • The balance increases by exactly the deposited amount.

REQ-ACC-04 Withdrawals and insufficient funds

Requirement	A customer can withdraw money from an active account, but never more than the available balance. Overdrafts are not offered.
Acceptance criteria	• A withdrawal larger than the balance is rejected with an insufficient-funds error. • After a rejected withdrawal, the balance is unchanged.

REQ-ACC-05 Account closure

Requirement	An account can be closed only when its balance is exactly SGD 0.00.
Acceptance criteria	• Closing an account with a non-zero balance is rejected, and the account stays ACTIVE. • Closing an account with a zero balance sets its status to CLOSED.

REQ-ACC-06 Frozen accounts

Requirement	An account in status FROZEN cannot receive deposits or make withdrawals.
Acceptance criteria	• A deposit to a frozen account is rejected. • A withdrawal from a frozen account is rejected. • The balance of a frozen account is not changed by rejected operations.

REQ-ACC-07 PIN lockout

Requirement	To prevent brute-force attacks, an account is automatically frozen after 3 consecutive failed PIN verification attempts.
Acceptance criteria	• After the third consecutive incorrect PIN, the account status becomes FROZEN. • A successful verification resets the failed-attempt counter to zero. • Two failures, then a success, then two more failures leaves the account ACTIVE.

REQ-ACC-08 Balance enquiry

Requirement	A balance enquiry returns the account balance rounded to two decimal places (half-up).
Acceptance criteria	• An account holding 10.005 reports a balance of 10.01.

4. Transfers module

REQ-TRF-01 Atomic funds transfer

Requirement	A transfer debits the source account and credits the destination account as one atomic operation.
Acceptance criteria	• After a completed transfer of X, the source balance is lower by X and the destination balance is higher by X. • If the credit cannot be applied (e.g. destination not active), the source balance is unchanged.

REQ-TRF-02 No self-transfers

Requirement	A transfer where source and destination are the same account is rejected.
Acceptance criteria	• Transferring from an account to itself raises a validation error.

REQ-TRF-03 Daily transfer limit

Requirement	The total of outgoing transfers from one account in a calendar day must not exceed SGD 5,000.00. The limit is inclusive : a day's total of exactly SGD 5,000.00 is allowed.
Acceptance criteria	• Given SGD 4,000.00 already transferred today (to an existing payee), a further transfer of SGD 1,000.00 is accepted. • After that, a transfer of SGD 0.01 is rejected with a daily-limit error. • Transfers made on a previous calendar day do not count towards today's limit.

REQ-TRF-04 New payee review

Requirement	A transfer of more than SGD 1,000.00 to a new payee is held for manual fraud review and not executed immediately.
Acceptance criteria	• A first-time transfer of SGD 1,000.01 or more to a payee has status PENDING_REVIEW, and neither balance changes. • A first-time transfer of exactly SGD 1,000.00 completes immediately. • Once one transfer to a payee has completed, later transfers to that payee are not held.

REQ-TRF-05 Transfer reference

Requirement	Every transfer gets a unique reference of the form "TRF-" followed by 8 upper-case alphanumeric characters, plus a timestamp.
Acceptance criteria	• References match TRF-[A-Z0-9]{8} and are unique. • Each transfer records the time it was created.

REQ-TRF-06 Transfer history

Requirement	A customer can view the transfer history of an account, including both incoming and outgoing transfers, most recent first.
Acceptance criteria	• History includes transfers where the account is the source or the destination. • Transfers are listed newest first. • Transfers between two other accounts do not appear.

REQ-TRF-07 Scheduled transfers

Requirement	A customer can schedule a transfer for a future date. The transfer runs automatically on that date and can be cancelled before then.
Acceptance criteria	• A scheduled transfer does not move funds before its execution date. • On the execution date, the transfer is executed under the same rules as REQ-TRF-01 to REQ-TRF-04. • A cancelled scheduled transfer is never executed.

5. Loans module

REQ-LN-01 Loan amount bounds

Requirement	Personal loans can be requested for SGD 1,000.00 to SGD 100,000.00 inclusive.
Acceptance criteria	• 999.99 and 100,000.01 are rejected; 1,000.00 and 100,000.00 are eligible.

REQ-LN-02 Loan tenure

Requirement	The loan tenure is 12 to 60 months inclusive.
Acceptance criteria	11 and 61 months are rejected; 12 and 60 months are eligible.

REQ-LN-03 Monthly instalment calculation

Requirement	The monthly instalment uses the standard amortisation formula $M = P \cdot r \cdot (1+r)^n / ((1+r)^n - 1)$, where r is the annual rate / 12 and n is the tenure in months. The result is rounded to the nearest cent (half-up) .
Acceptance criteria	- SGD 25,000.00 at 5.5% p.a. over 36 months → instalment SGD 754.90. - SGD 60,000.00 at 4.5% p.a. over 48 months → instalment SGD 1,368.21.

REQ-LN-04 Affordability check

Requirement	An application is rejected when the applicant's monthly income is less than 3 times the monthly instalment.
Acceptance criteria	- Income below 3× the instalment → REJECTED with an affordability reason. - Income of 3× the instalment or more → eligible.

REQ-LN-05 Interest rate tiers

Loan amount	Annual rate
Below SGD 20,000.00	6.5%
SGD 20,000.00 to SGD 50,000.00 inclusive	5.5%
Above SGD 50,000.00	4.5%

REQ-LN-06 Early repayment fee

Requirement	A borrower who repays early pays a fee of 1.5% of the outstanding principal, rounded to the nearest cent (half-up).
Acceptance criteria	- Outstanding principal SGD 12,345.67 → fee SGD 185.19. - Outstanding principal SGD 10,000.00 → fee SGD 150.00.

REQ-LN-07 Loan decision

Requirement	Every loan application gets a decision of APPROVED or REJECTED. Rejected applications carry a human-readable reason.
Acceptance criteria	• Approved loans have no rejection reason; rejected loans always have one. • A decided application can be retrieved by its identifier.

6. Security

These requirements apply to the customer-facing API. They supplement NBK-SEC-2025-021 (Authentication standard).

REQ-SEC-01 PIN protection

Requirement	Customer PINs are stored only as a salted one-way hash, never in plain text, and are never returned by the API.
Acceptance criteria	• The stored PIN value for an account is not the plain-text PIN. • The correct PIN verifies successfully; an incorrect PIN does not. • Account data returned by the API contains no PIN or PIN hash field.

REQ-SEC-02 Session tokens

Requirement	A successful login issues a signed session token that identifies the customer's account and expires 60 minutes after it is issued.
Acceptance criteria	• A freshly issued token is accepted and identifies the account it was issued for. • A token whose content or signature has been altered is rejected. • An expired token is rejected. • The token lifetime is 60 minutes.

REQ-SEC-03 Login throttling

Requirement	After 5 failed login attempts for an account within 15 minutes, further login attempts for that account are refused until the window has passed. This temporary throttle is separate from the permanent freeze in REQ-ACC-07.
Acceptance criteria	• After 5 failed attempts, the next attempt for that account is refused, even with the correct PIN. • 4 failed attempts do not block the account. • A successful login resets the failed-attempt count. • Failed attempts on one account do not affect other accounts.

REQ-SEC-04 Audit trail

Requirement	Security-relevant events are recorded in an append-only audit log with a timestamp, the account number and the event type.
Acceptance criteria	• Successful and failed logins are recorded. • Deposits, withdrawals (including rejected withdrawals) and completed transfers are recorded. • Account freezes and closures are recorded. • Audit entries cannot be modified or removed through the audit log's interface.

REQ-SEC-05 Account ownership

Requirement	A logged-in customer may only act on their own account. Viewing account details or balance, withdrawing, closing, viewing transfer history and sending a transfer from an account are only allowed for the customer logged in to that account.
Acceptance criteria	• A customer requesting another customer's account details is refused with HTTP 403. • A customer attempting a transfer from another customer's account is refused with HTTP 403, and no money moves. • A customer can view their own account details. • Requests without a valid session token are refused.

7. Change control

Changes to this document need Product Office approval. Engineering must re-establish traceability for every changed requirement before the next release.